Writing the Jetson Clean System

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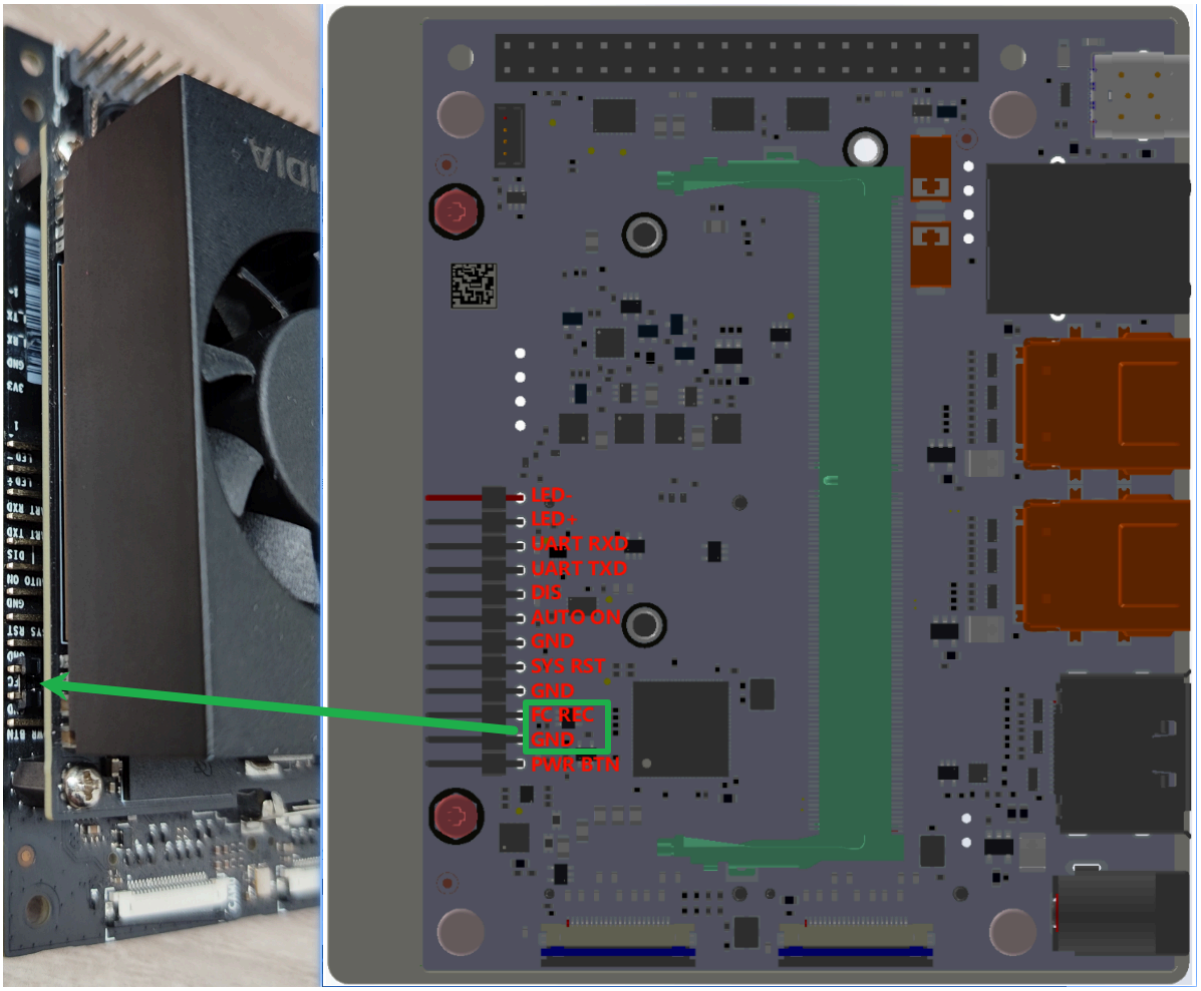
The tutorial demonstrates writing the NVIDIA official system image on the Jetson Orin **official kit** (the SUB kit cannot activate SUPER mode in this way). Beginners are not advised to perform this operation!

Note: The SSD that comes with the product contains a factory image, and the factory image includes the NVIDIA official system and the mainboard environment; if you do not require a clean system, do not operate!

1. Hardware Connection

1.1. Recovery Mode

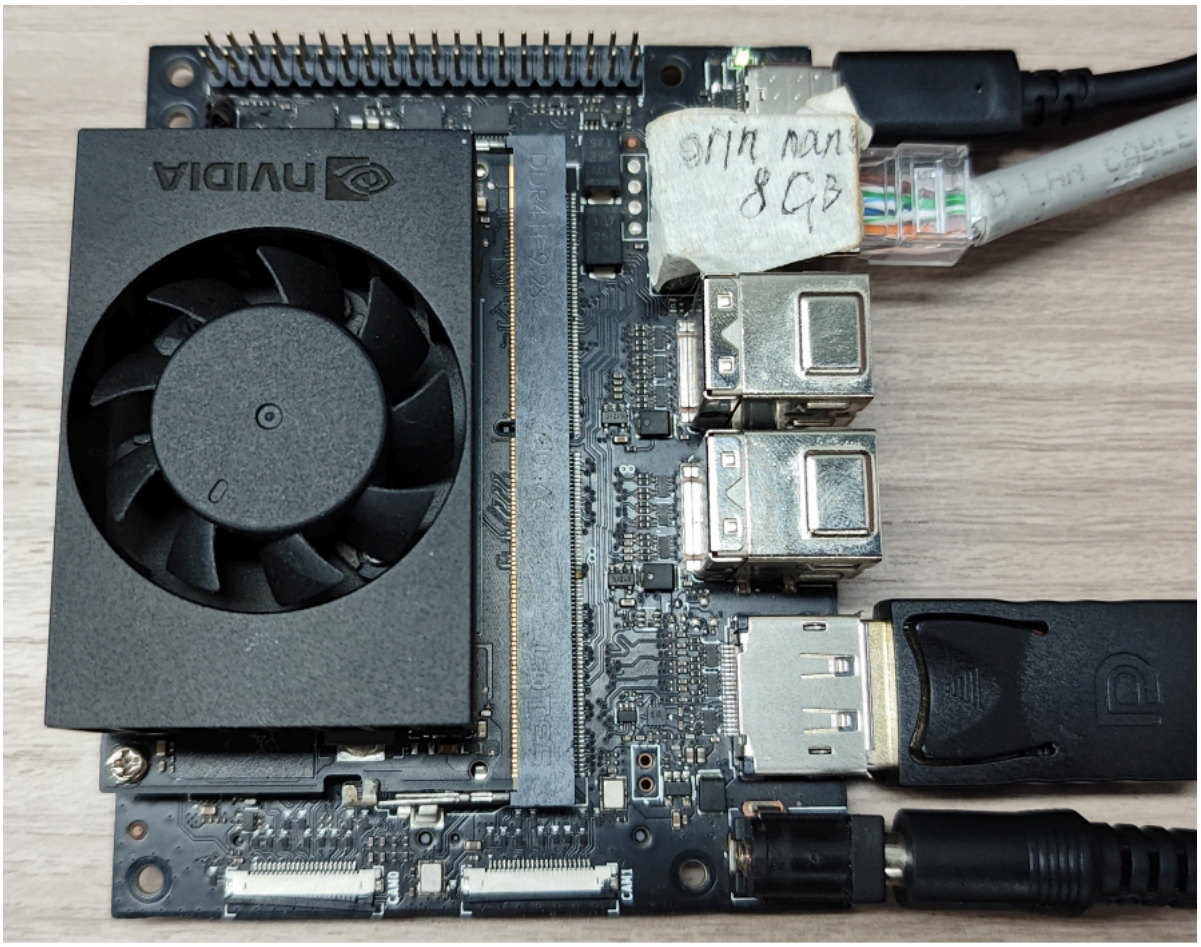
The Jetson Orin mainboard needs to short `FC REC` and `GND` under the core board to enter Recovery mode:



1.2. Mainboard Connection

This illustration is based on the official version of Jetson Orin Nano. Users of other versions can refer to it (the hardware interfaces and functional layout are the same)

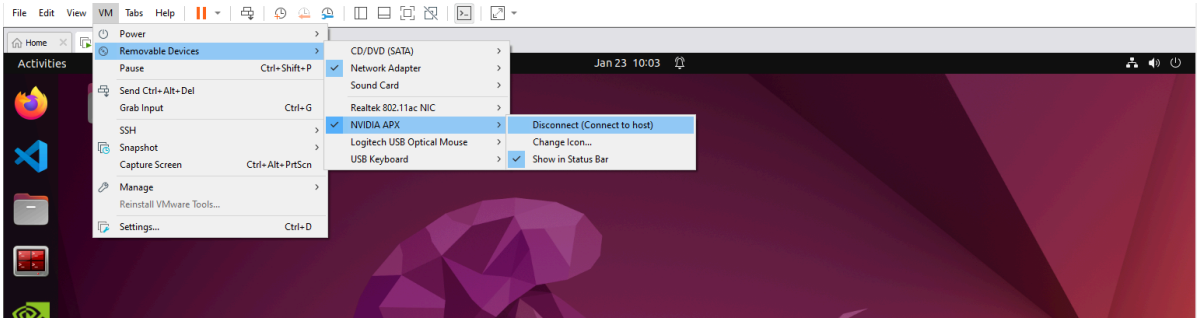
The Jetson Orin mainboard needs to be connected to a DC power adapter, a DP data cable, a network cable, and a Type C data cable: the Type C data cable connects to the computer



2. Software Connection

The tutorial uses VMware Workstation 17 Pro to write the system!

After starting the virtual machine, select in the menu bar: Virtual Machine → NVIDIA APX → Confirm the connection status with the virtual machine



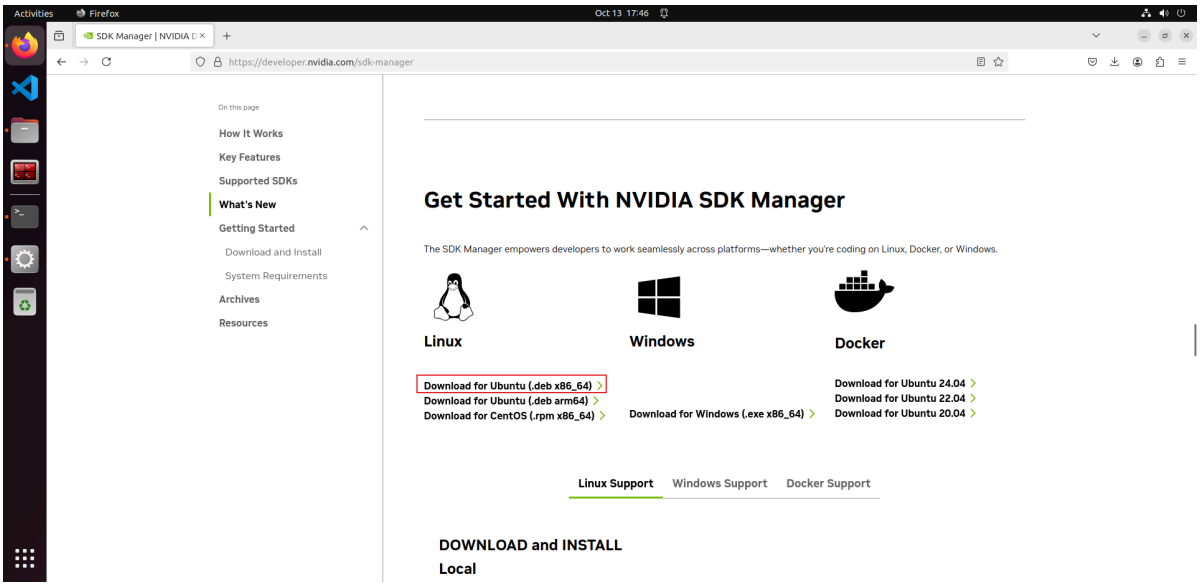
3. Using SDK Manager

Using SDK Manager requires an account registration and login in advance; complete the account registration and login by yourself!

```
No need to install SDK Manager when using our virtual machine. If SDK Manager cannot be updated, reinstall it according to the tutorial below
```

3.1. SDK Manager Download

Official website: https://developer.nvidia.com/sdk-manager



> Note: The official website may update the version. If a new version is downloaded from the official website during your own build, modify the command below accordingly. To use the same version as the tutorial, visit the official historical version summary website: https://developer.nvidia.com/nvidia-sdk-manager-archive, or use the installation package provided in the cloud drive.

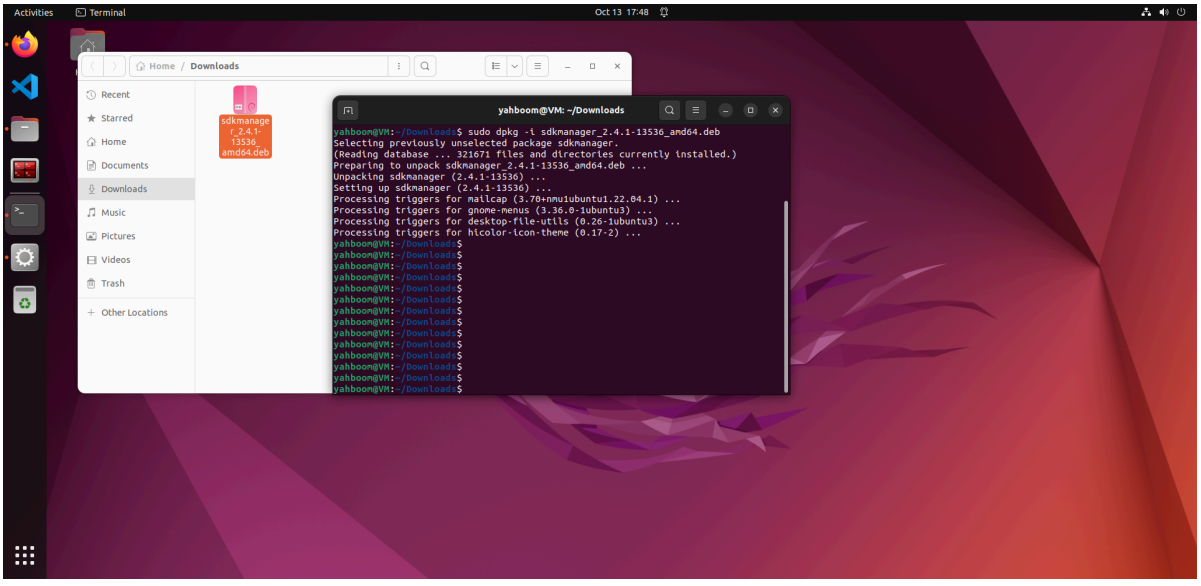
3.2. SDK Manager Installation

Enter the folder where the installation package is located and run the installation command. Note that the file name in the command must match the actual installation package:

```
sudo dpkg -i sdkmanager_2.4.1-13536_amd64.deb
```

Repair dependencies: A clean system may show an error message when installing sdkmanager. Run the command below, then run the installation command again

```
sudo apt --fix-broken install -y
```

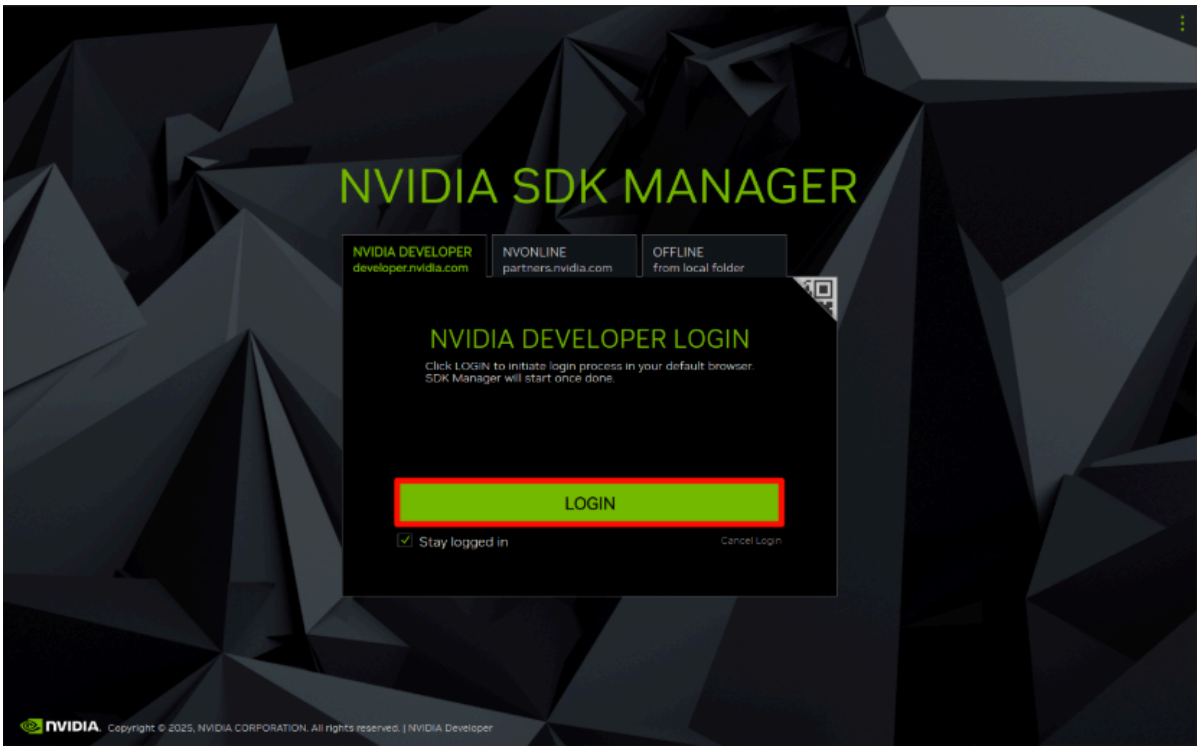


3.3. Starting and Logging in to the Account

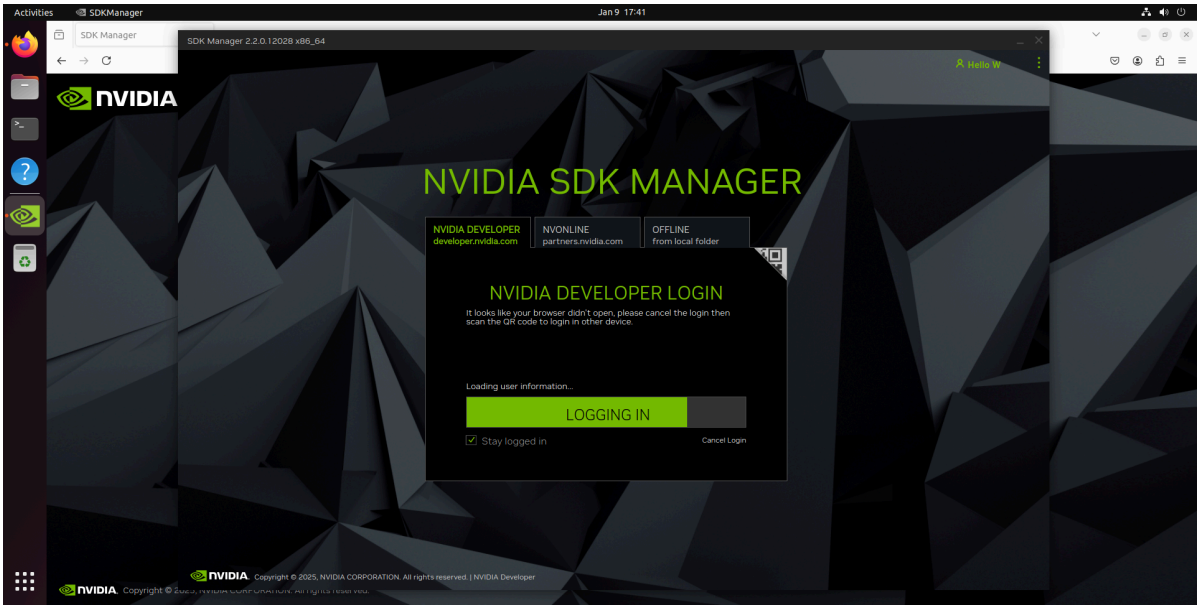
Find the `SDK Manager` application icon in the system application menu bar, or enter the command below in the terminal to start it:

```
sdkmanager
# To write an old version of Jetpack, use the command: sdkmanager --archived-versions
```

After clicking login, the browser login page pops up automatically:

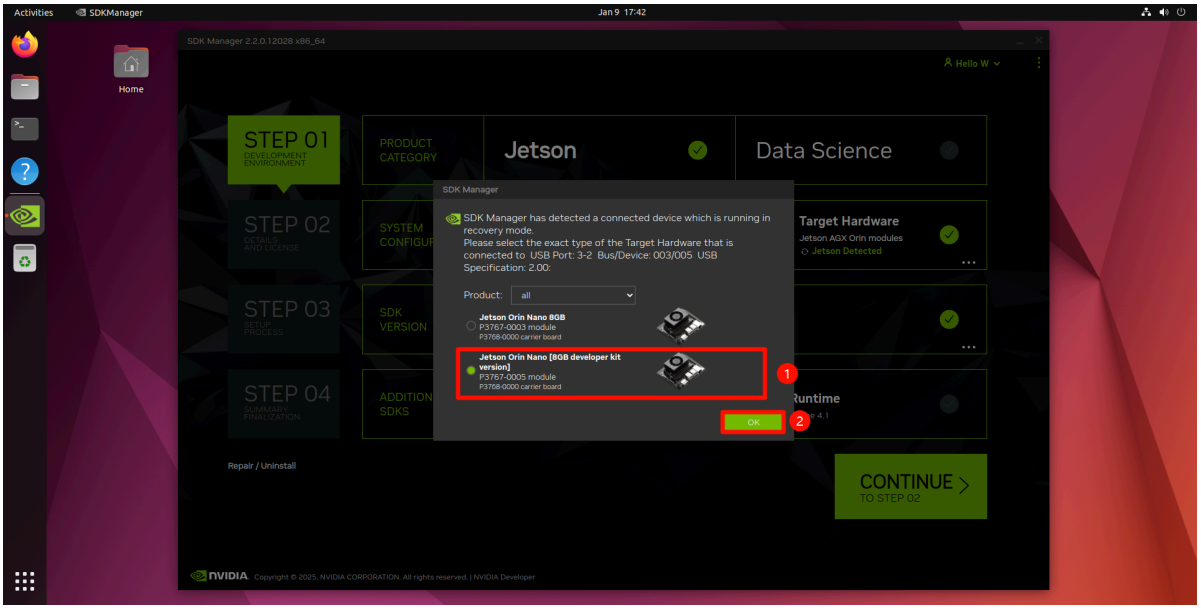


Enter the previously registered account and password to enter the software:



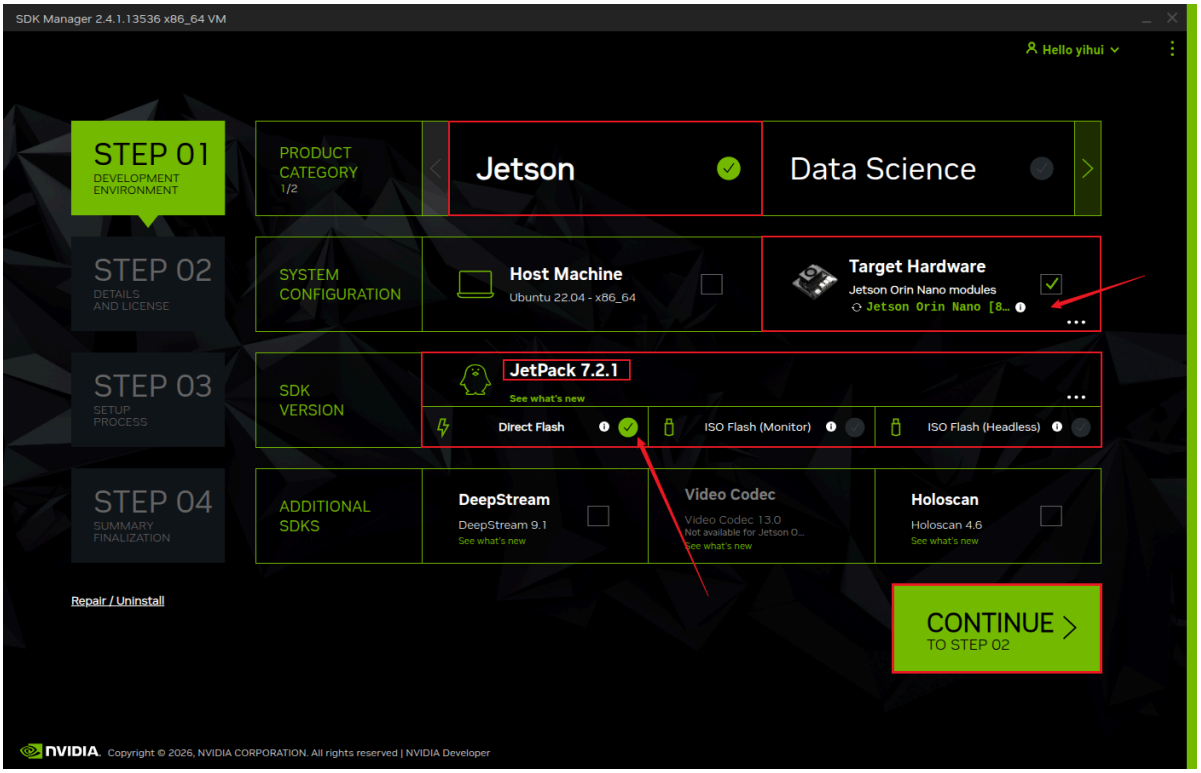
3.4. Mainboard Selection

Select according to your own mainboard model: for the official kit, select `Jetson Orin Nano [*GB developer kit version]`



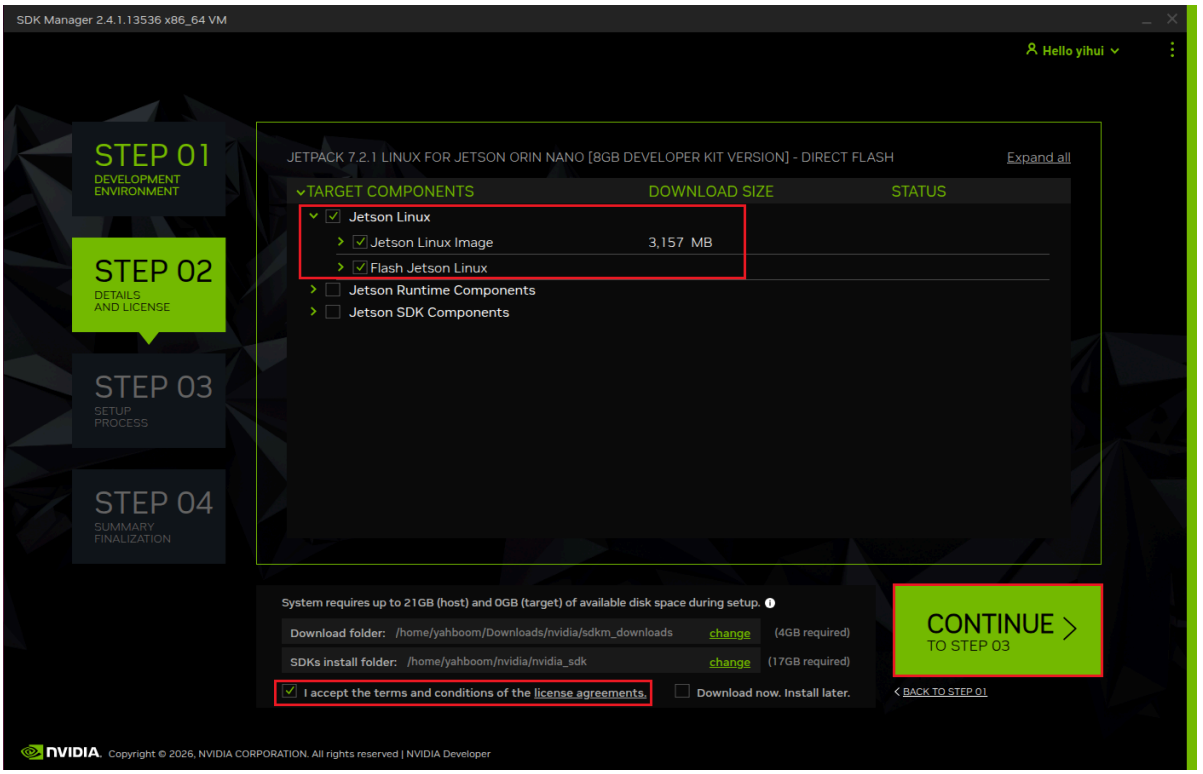
3.5. STEP1

Check the options below, and click `CONTINUE` after confirming that the selections are consistent:

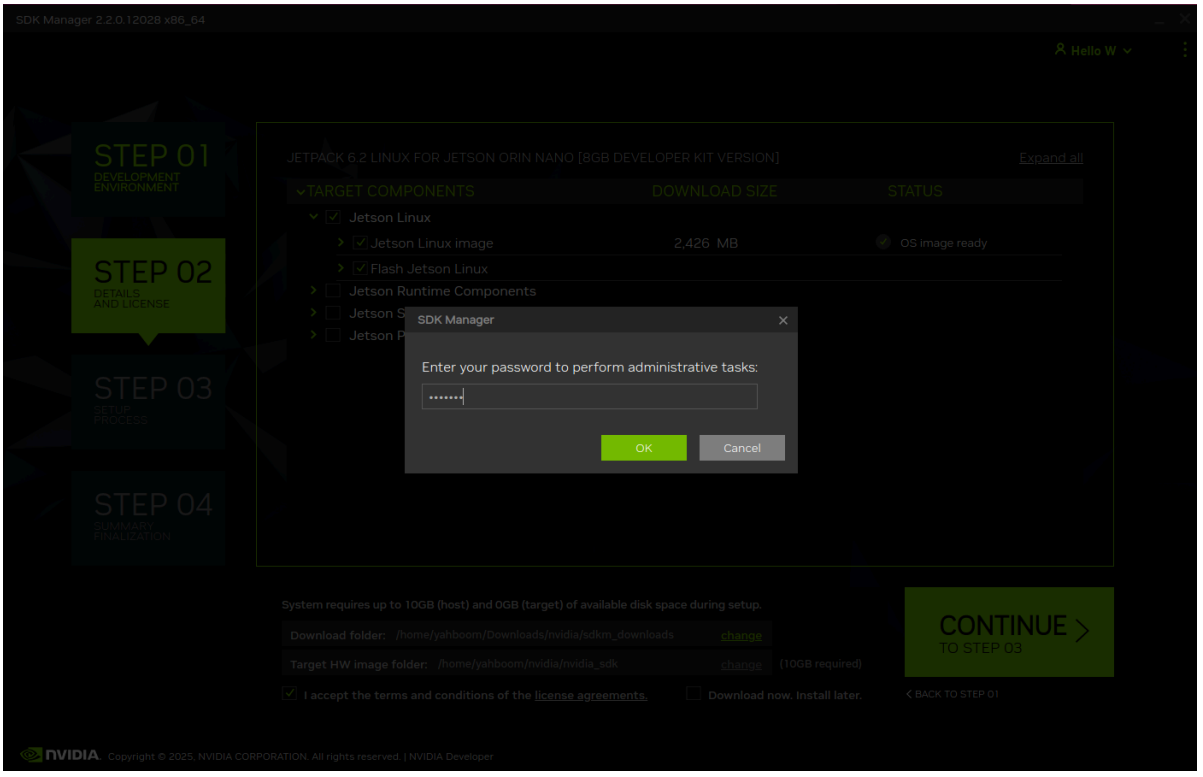


3.6. STEP2

Check the options below, and click `CONTINUE` after confirming that the selections are consistent:



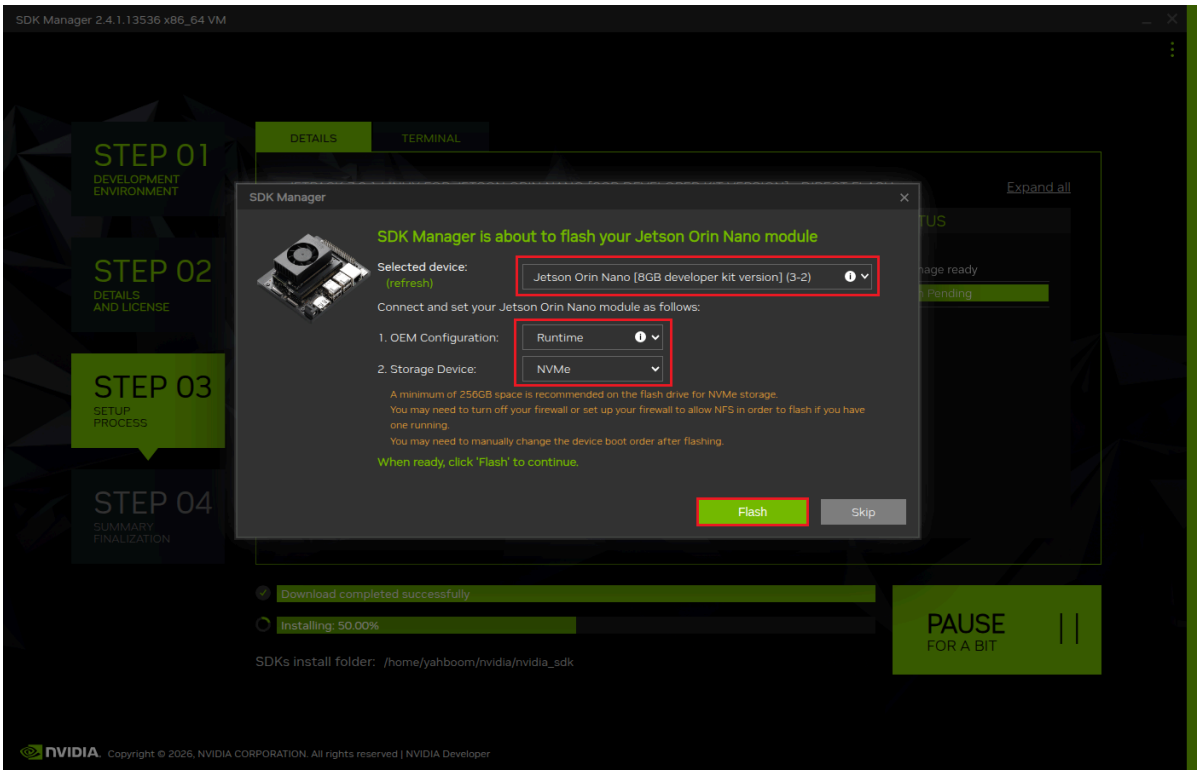
Enter the virtual machine password: yahboom. If it is a personal virtual machine, enter your own password



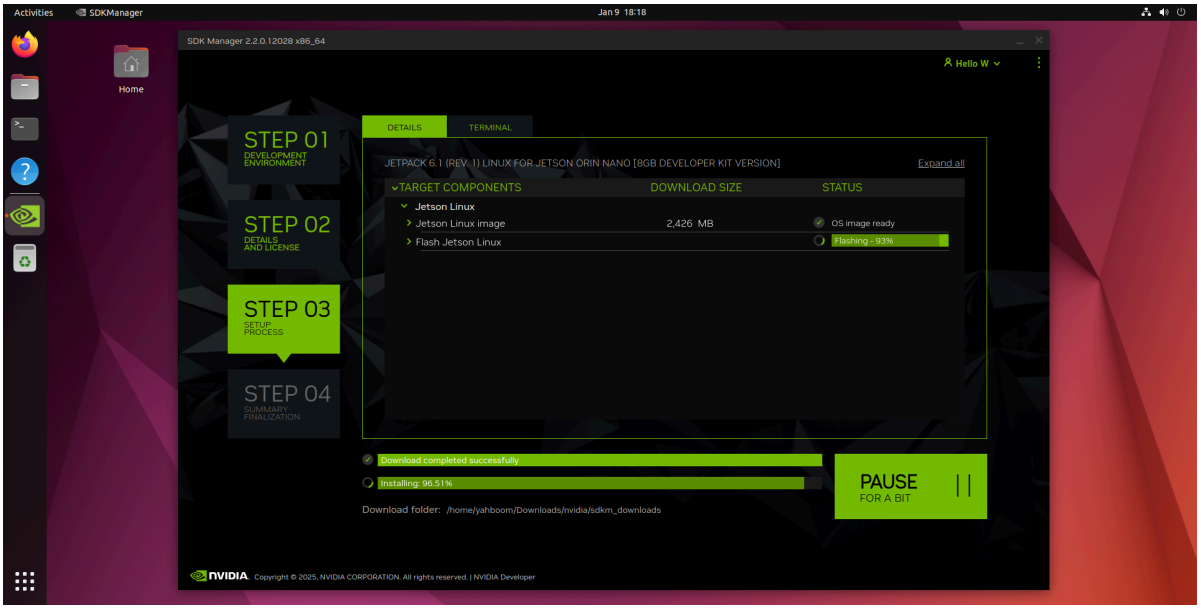
3.7. STEP3

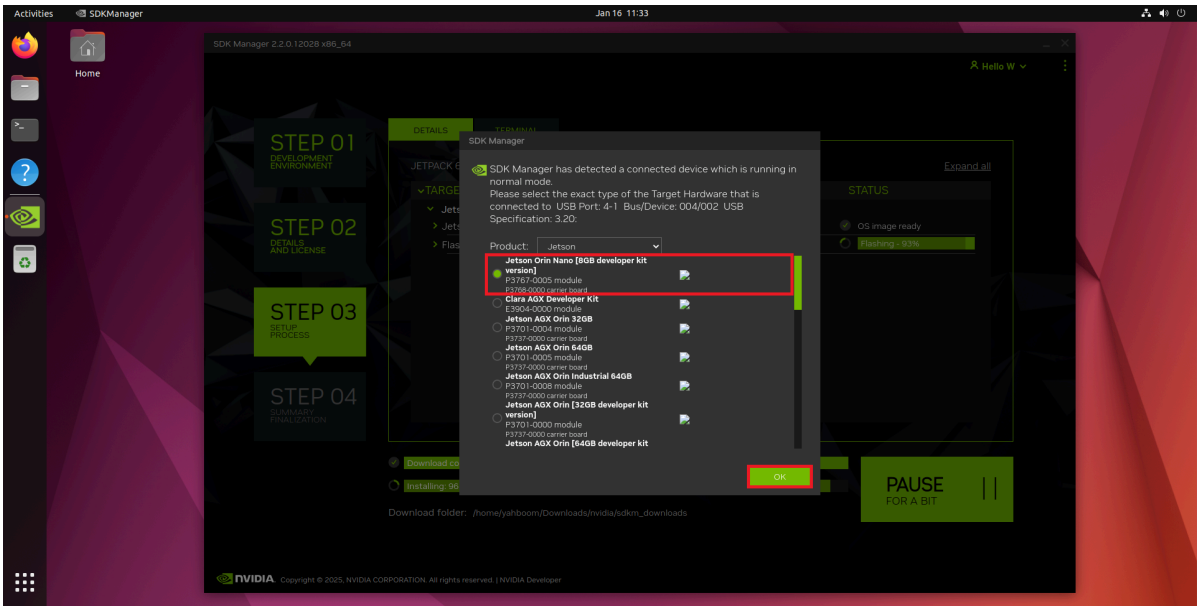
Wait for the system to download and install: the user needs to set options when the system installation reaches 50%

Select the mainboard model, OEM configuration, and system storage medium:

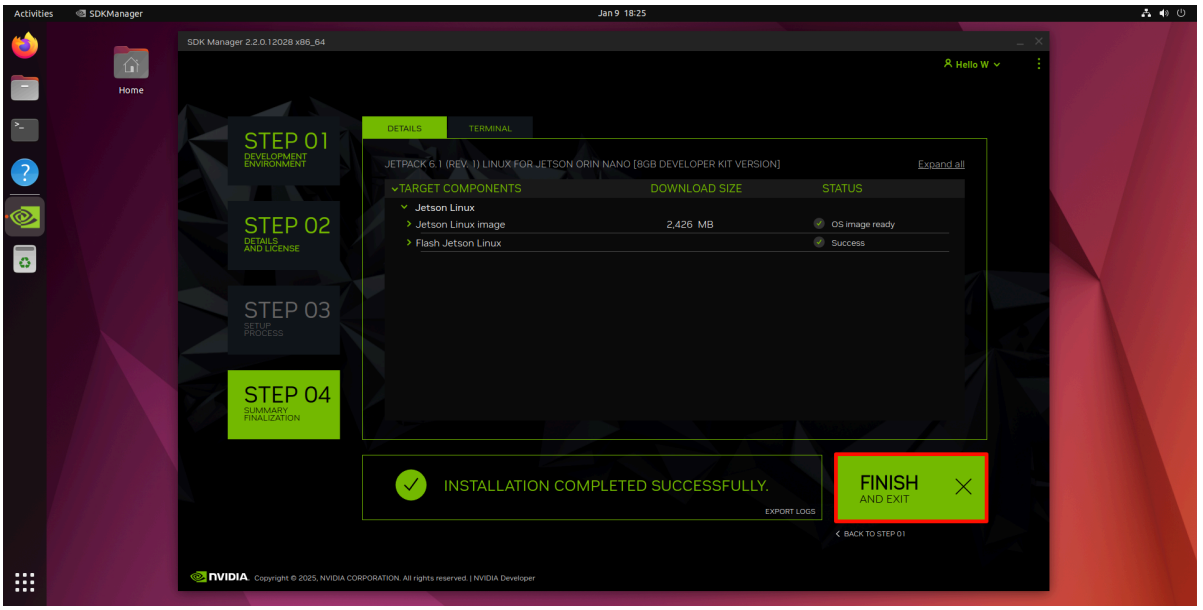


The virtual machine automatically controls the connection and disconnection of the device during the whole process, but once the progress exceeds 9%, pay attention to the SDK Manager system writing progress. If the device is disconnected and does not automatically connect to the virtual machine, manually connect the mainboard to the virtual machine in time:





3.8. STEP4



4. Starting the System

After successfully writing the system with SDK Manager, disconnect the DC power supply and the Type C data cable of the mainboard. With the power off, remove the jumper shorting the FC REC and GND pins under the core board, then connect a monitor, keyboard, and mouse to the mainboard for the first system setup, and finally power the mainboard with the DC power adapter!

4.1. Normal Mode

Unplug the `FC REC` and `GND` jumper caps shorted when writing the Jetson Orin mainboard.

4.2. Starting the System

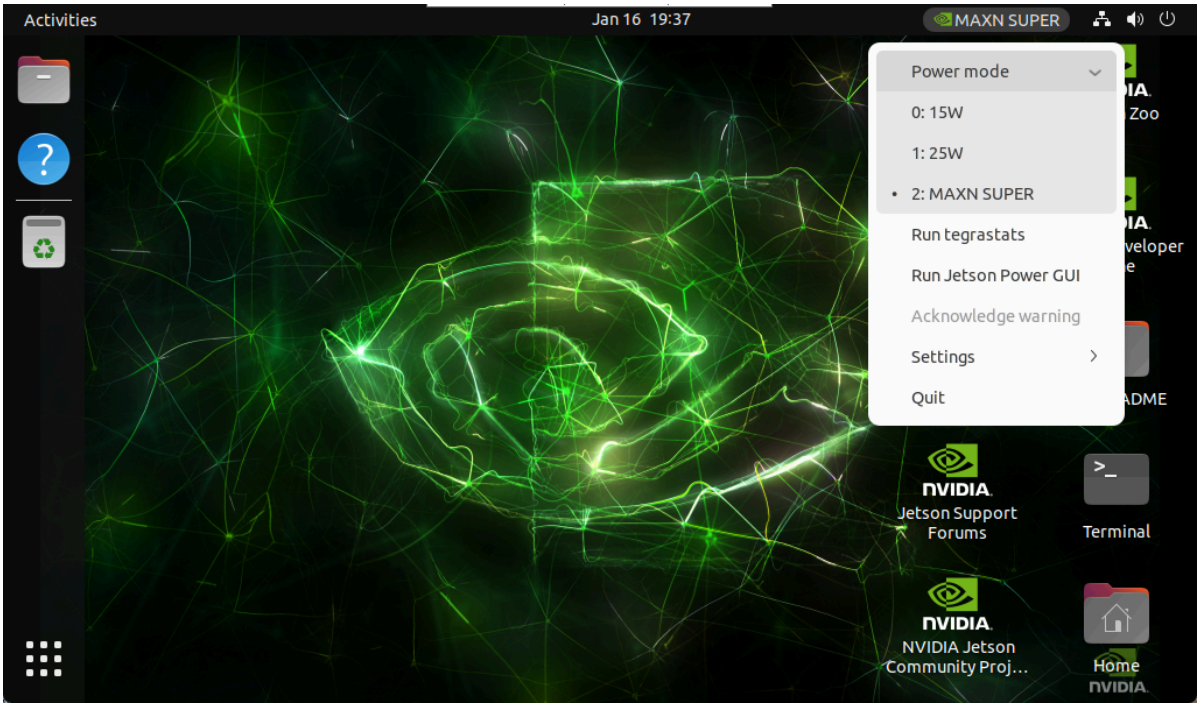
Connect the DP data cable, network cable, and Type C data cable to the Jetson mainboard, and finally power the Jetson mainboard with the DC power adapter.

```
The Jetson Orin series mainboard needs a monitor connected to display the screen normally and to perform VNC remote access
```

4.3. System Settings

After starting the system, users can set the system options according to the system prompts. The process is relatively simple and is not demonstrated here. Finally, enter the system:





Frequently Asked Questions

Unrecognized Device

Disconnect the DC power supply and the Type C data cable of the mainboard, then power it on again and connect it to the virtual machine!